CSCI-570 Homework 5

Due Date: Tuesday, November 4th, 11:59pm

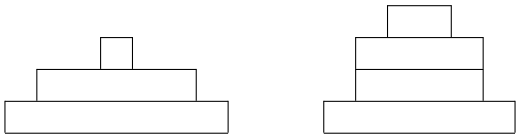
Problem 1.
The dynamic programming algorithm that finds the optimal solution of the *Weighted Interval Scheduling* problem (Kleinberg-Tardos, pp. 252–260) presupposes that the finish times of the intervals are ordered non-decreasingly, i.e. $f_i \leq f_j$ in case of $i < j$ and that the function $p(j)$, which for interval j returns the largest index $i < j$ such that intervals i and j are disjoint, is pre-calculated. Show that the ordering of the finish times of the intervals and the calculation of the values of $p(j)$ can be done in $O(n \cdot log\ n)$ time.

[6 points]

Problem 2.
We are building towers via stacking building blocks on top of each other. A tower is called *stable* if no larger block is placed over a smaller one (see the diagram below). Given m different sizes of building blocks with k pieces of each sizes, how many different stable towers can be built of height n? (We consider two towers to be different, if at least on one level $1 \leq l \leq n$ they contain different sizes of building blocks.)

Given three positive integers n, m and k as input, provide a dynamic programming algorithm that calculates the number of different stable towers of height n, while using m different sizes of building blocks with k copies each. Is the runtime of the algorithm polynomial?

[Hint: you don't need a 3D array, you can include the parameter k in the recurrence formula.]



Stable towers of height 3 and 4.

Examples:
Input: $n = 3$, $m = 3$, $k = 1$. Output: 1. Stable towers: $[1, 2, 3]$.
Input: $n = 3$, $m = 3$, $k = 2$. Output: 7.
Stable towers: $[1, 1, 2], [1, 1, 3], [1, 2, 2], [1, 2, 3], [1, 3, 3], [2, 2, 3], [2, 3, 3]$.

[8 points]

Problem 3.
Suppose you want to drive from USC to Santa Monica. Your gas tank, when full, holds enough gas for m miles. Suppose there are n gas stations along the route at distances $d_1 \leq d_2 \leq ... \leq d_n$ from USC. Assume that the distance between any neighboring gas stations, and the distance between USC and the first gas station, as well as the distance between the last gas station and Santa Monica are all at most m miles. Assume you start from USC with the tank full. Your goal is to stop at as few gas stations along the way as possible. Give the most efficient algorithm to determine which gas stations

you should stop at and prove that your algorithm yields an optimal solution (i.e., the minimum number of gas stops). Give the time complexity of your algorithm as a function of n.

[8 points]

Problem 4.
You are given a binary tree $T(V, E)$. Every node x is in *conflict* with all its neighbors (i.e. with its left child, right child and parent, if they exist). Our goal is to find the largest subset of non-conflicting nodes. Consider the following algorithm:

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while V is not empty
  add a leaf node x ∈ V to S
  remove x and all nodes that conflict with it from V
return S
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Thus, S is the set of non-conflicting nodes our greedy algorithm builds. Recognize that while removing x and all its neighbors may disconnect the tree, it does not prevent continuing to choose leaf nodes from each disconnected tree until V is exhausted.

Give a rigorous proof of the correctness of this algorithm.

[8 points]